

6. Develop a Student Result Management System using JSP, JDBC, JSTL, SQL, and Exception Handling to perform database-driven CRUD operations.

Index.html

```
<!DOCTYPE html>

<html>

  <head>

    <title>Student Result Management</title>

  </head>

  <body>

    <h1>Enter Student Details</h1>

    <form action="StudentStoreResult.jsp" method="post">

      Id: <input type="number" name="id"><br>

      Name: <input type="text" name="name"><br>

      Course: <input type="text" name="course"><br>

      Marks: <input type="number" name="marks"><br>

      <button type="submit">Submit</button>

    </form>

    <a href="ViewResult.jsp"> View Results </a>

    <a href="UpdateResult.jsp"> Update Results </a>

    <a href="DeleteResult.jsp"> Delete Results </a>

  </body>

</html>
```

StudentStoreResult.jsp

```
<%@page contentType="text/html" pageEncoding="UTF-8"%>

<%@taglib prefix="sql" uri="jakarta.tags.sql" %>

<%@taglib prefix="c" uri="jakarta.tags.core" %>

<sql:setDataSource var="db" driver="com.mysql.cj.jdbc.Driver"
```

```
url="jdbc:mysql://localhost:3306/students" user="root"
password="root" />
```

```
<c:if test="${pageContext.request.method == 'POST'}">
```

```
<sql:update dataSource="${db}" var="status">
```

```
insert into result values (?, ?, ?, ?);
```

```
<sql:param value="${param.id}"/>
```

```
<sql:param value="${param.name}"/>
```

```
<sql:param value="${param.course}"/>
```

```
<sql:param value="${param.marks}"/>
```

```
</sql:update>
```

```
</c:if>
```

```
<c:if test="${status>0}">
```

```
<h1> Marks added Successfully. </h1>
```

```
</c:if>
```

```
<c:if test="${status==0}">
```

```
<h1> Failed to add marks. </h1>
```

```
</c:if>
```

ViewResult.jsp

```
<%@page contentType="text/html" pageEncoding="UTF-8"%>
```

```
<%@taglib prefix="sql" uri="jakarta.tags.sql" %>
```

```
<%@taglib prefix="c" uri="jakarta.tags.core" %>
```

```
<sql:setDataSource var="db" driver="com.mysql.cj.jdbc.Driver"
```

```
url="jdbc:mysql://localhost:3306/students" user="root"
```

```
password="root" />
```

```

<sql:query dataSource="${db}" var="result">
    select * from result;
</sql:query>

<c:forEach var="res" items="${result.rows}">
    <h1> ${res.id} | ${res.name} | ${res.course} | ${res.marks} </h1>
</c:forEach>

```

UpdateResult.jsp

```

<%@page contentType="text/html" pageEncoding="UTF-8"%>
<%@taglib prefix="sql" uri="jakarta.tags.sql" %>
<%@taglib prefix="c" uri="jakarta.tags.core" %>

<sql:setDataSource var="db" driver="com.mysql.cj.jdbc.Driver"
    url="jdbc:mysql://localhost:3306/students" user="root"
    password="root" />

<!DOCTYPE html>

<html>

<head>
    <title>Update Page</title>
</head>

<body>
    <h1>Update Student Result</h1>
    <form action="UpdateResult.jsp" method="post">
        Existing Id: <input type="number" name="id"><br>
        Updated Name: <input type="text" name="name"><br>
        Updated Course: <input type="text" name="course"><br>
        Updated Marks: <input type="number" name="marks"><br>
    </form>

```

```

        <button type="submit">Submit</button>

    </form>

    <a href="ViewResult.jsp"> View Results </a>

</body>
</html>

<c:if test="${pageContext.request.method == 'POST'}">
<sql:update dataSource="${db}" var="status">
    update result set name=?, course=?, marks =? where id=?;

    <sql:param value="${param.name}"/>
    <sql:param value="${param.course}"/>
    <sql:param value="${param.marks}"/>
    <sql:param value="${param.id}"/>
</sql:update>
</c:if>

    <c:if test="${status>0}">
        <h1> Marks updated Successfully. </h1>
    </c:if>

    <c:if test="${status==0}">
        <h1> Failed to update marks. </h1>
    </c:if>

```

DeleteResult.jsp

```

<%@page contentType="text/html" pageEncoding="UTF-8"%>
<%@taglib prefix="sql" uri="jakarta.tags.sql" %>
<%@taglib prefix="c" uri="jakarta.tags.core" %>

<sql:setDataSource var="db" driver="com.mysql.cj.jdbc.Driver"

```

```
url="jdbc:mysql://localhost:3306/students" user="root"
password="root" />
```

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
<title>Delete Page</title>
```

```
</head>
```

```
<body>
```

```
<h1>Delete Student Result</h1>
```

```
<form action="DeleteResult.jsp" method="post">
```

```
Existing Id: <input type="number" name="id"><br>
```

```
<button type="submit">Submit</button>
```

```
</form>
```

```
<a href="ViewResult.jsp"> View Results </a>
```

```
</body>
```

```
</html>
```

```
<c:if test="${pageContext.request.method == 'POST'}">
```

```
<sql:update dataSource="${db}" var="status">
```

```
delete from result where id=?;
```

```
<sql:param value="${param.id}"/>
```

```
</sql:update>
```

```
</c:if>
```

```
<c:if test="${status>0}">
```

```
<h1> Result deleted Successfully. </h1>
```

```
</c:if>
```

```
<c:if test="${status==0}">
```

<h1> Failed to delete result. </h1>

</c:if>